Linear Algebra | **Unit 4: Orthogonality**

Practice Test — Study Copy

Name: ____________________________ Date: ________________ Period: _________

Keep in Mind

This is a practice test. It mirrors the real Unit 4 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 pts)

Write the letter of the best description next to each term.

1. Orthonormal vectors _______
2. Orthogonal matrix Q _______
3. Projection matrix P _______
4. Normal equations _______
5. Least squares _______
6. Residual (error) _______
7. Orthogonal complement _______
8. Gram–Schmidt _______

(A) The system $A^{\mathsf{T}}A\,\hat{\mathbf{x}} = A^{\mathsf{T}}\mathbf{b}$ whose solution is the best $\hat{\mathbf{x}}$.

(B) The process that turns independent vectors into an orthonormal set (and builds $A = QR$).

(C) Perpendicular *unit* vectors: $q_i \cdot q_j = 0$ and each $q_i \cdot q_i = 1$.

(D) The error vector $\mathbf{e} = \mathbf{b} - \mathbf{p}$; it lies in $N(A^{\mathsf{T}})$, perpendicular to $C(A)$.

(E) A square matrix with orthonormal columns, so $Q^{\mathsf{T}}Q = I$ and $Q^{\mathsf{T}} = Q^{-1}$.

(F) All vectors perpendicular to a subspace V; together with V it fills the whole space.

(G) Choosing $\hat{\mathbf{x}}$ to make the error $\|\mathbf{b} - A\mathbf{x}\|^2$ as small as possible.

(H) The matrix $P = \dfrac{\mathbf{a}\mathbf{a}^{\mathsf{T}}}{\mathbf{a}^{\mathsf{T}}\mathbf{a}}$ that sends $\mathbf{b}$ to its projection $\mathbf{p} = P\mathbf{b}$.

Part B — Multiple Choice (12 pts)

Circle the best answer.

1. The least-squares solution makes $A\hat{\mathbf{x}}$ the point of $C(A)$ that is (A) farthest from $\mathbf{b}$ (B) closest to $\mathbf{b}$ (C) equal to $\mathbf{b}$ (D) zero
2. For an orthogonal matrix Q, the product $Q^{\mathsf{T}}Q$ equals (A) 0 (B) Q (C) I (D) Q^2
3. The least-squares error $\mathbf{e} = \mathbf{b} - \mathbf{p}$ is perpendicular to (A) every column of A (B) $\hat{\mathbf{x}}$ (C) nothing (D) $\mathbf{b}$
4. The normal equations for the best solution of $A\mathbf{x} = \mathbf{b}$ are (A) $A\mathbf{x} = \mathbf{b}$ (B) $A^{\mathsf{T}}A\,\hat{\mathbf{x}} = A^{\mathsf{T}}\mathbf{b}$ (C) $AA^{\mathsf{T}}\mathbf{x} = \mathbf{b}$ (D) $\hat{\mathbf{x}} = \mathbf{b}$
5. If Q has orthonormal columns, the coordinates of $\mathbf{b}$ in that basis are found by (A) solving a big system (B) the dot products $q_i \cdot \mathbf{b}$ (C) inverting $A^{\mathsf{T}}A$ (D) guessing
6. A projection matrix P always satisfies (A) $P^2 = P$ (B) $P^2 = I$ (C) $P = 0$ (D) $P^{\mathsf{T}} = -P$

Part C — Short Answer & Computation (35 pts)

Show your work.

1. Let V be the line through $\mathbf{a} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$ in $\mathbb{R}^3$. Decide whether each vector is in the orthogonal complement $V^\perp$ (show the dot product): **(a)** $\mathbf{w} = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}$ **(b)** $\mathbf{u} = \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}$.

2. Project $\mathbf{b} = \left[\begin{smallmatrix} 3 \\ 1 \end{smallmatrix}\right]$ onto the line through $\mathbf{a} = \left[\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right]$. Find $\hat{x} = \dfrac{\mathbf{a} \cdot \mathbf{b}}{\mathbf{a} \cdot \mathbf{a}}$, the projection $\mathbf{p} = \hat{x}\mathbf{a}$, and the error $\mathbf{e} = \mathbf{b} - \mathbf{p}$. Verify that $\mathbf{e} \perp \mathbf{a}$.

3. Use the **normal equations** $A^\mathsf{T} A \hat{\mathbf{x}} = A^\mathsf{T}\mathbf{b}$ to project $\mathbf{b} = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix}$ onto $C(A)$, where $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$. Find $\hat{\mathbf{x}}$, $\mathbf{p} = A\hat{\mathbf{x}}$, and $\mathbf{e} = \mathbf{b} - \mathbf{p}$; check that $A^\mathsf{T}\mathbf{e} = \mathbf{0}$.

4. Fit the best line $y = C + Dt$ to the data $(t, y) = (0, 3), (1, 3), (2, 5), (3, 9)$ by least squares. Set up A and $\mathbf{b}$, solve the normal equations for (C, D), and list the residuals $\mathbf{e} = \mathbf{b} - \mathbf{p}$. What is $\sum e_i$?

5. Let $q_1 = \frac{1}{5}\left[\begin{smallmatrix} 3 \\ 4 \end{smallmatrix}\right]$ and $q_2 = \frac{1}{5}\left[\begin{smallmatrix} 4 \\ -3 \end{smallmatrix}\right]$. Show they are **orthonormal** (check $q_1 \cdot q_1$, $q_2 \cdot q_2$, and $q_1 \cdot q_2$), then explain why the matrix $Q = \begin{bmatrix} q_1 & q_2 \end{bmatrix}$ satisfies $Q^\mathsf{T} Q = I$.

6. Using the same q_1, q_2 as above, write $\mathbf{b} = \left[\begin{smallmatrix} 5 \\ 0 \end{smallmatrix}\right]$ in the orthonormal basis — i.e. find c_1, c_2 with $\mathbf{b} = c_1 q_1 + c_2 q_2$ — *without* solving a system. (Hint: $c_i = q_i \cdot \mathbf{b}$.)

7. Run **Gram–Schmidt** on $\mathbf{a}_1 = \left[\begin{smallmatrix} 3 \\ 4 \end{smallmatrix}\right]$, $\mathbf{a}_2 = \left[\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right]$ to produce an orthonormal pair q_1, q_2. (Normalize $\mathbf{a}_1$; subtract its

part from $\mathbf{a}_2$; normalize the remainder.)

Part D — Extended Response (10 pts)

Explain your reasoning in full sentences.

1. A least-squares line fit produces fitted values $\mathbf{p} = A\hat{\mathbf{x}}$ and residuals $\mathbf{e} = \mathbf{b} - \mathbf{p}$. **(a)** Explain why "best fit" forces the error $\mathbf{e}$ to be perpendicular to the column space $C(A)$, so that $A^{\mathsf{T}}\mathbf{e} = \mathbf{0}$. **(b)** When A has an all-ones column, explain why this makes $\sum e_i = 0$ — the positive and negative residuals must cancel.

2. Let Q be a square matrix with orthonormal columns. **(a)** Explain why the coordinates of any $\mathbf{b}$ are just the dot products $q_i \cdot \mathbf{b}$, so $\mathbf{b} = \sum (q_i \cdot \mathbf{b})\, q_i$ with no system to solve. **(b)** Explain why least squares collapses to $\hat{\mathbf{x}} = Q^{\mathsf{T}}\mathbf{b}$ when $A = Q$. Use the fact that $Q^{\mathsf{T}}Q = I$.